Student Club Management System — The Full Documentation

Complete System Requirements, Governance Rules & Platform Specification

AUDIENCE	STATUS	DOCUMENT INDEX
Engineering Team	In Review	System Specification

Table Of Contents

System Governance & Platform

A. System Authorization & Governance

- a. Roles Glossary
- b. Role Hierarchy
- c. Visibility
- e. Permissions

B. Platform & Navigation

- B.1 Mobile Platform
- B.2 Web Platform

Interfaces, Journeys & Rules

C. Screens

D. Per-Role User Journeys

- D.1 Guest to User Journey
- D.2 Student Journey
- D.3 Club President Journey
- D.4 Assistant Member Journey
- D.5 Club Member Journey
- D.6 Coordination Club President Journey
- D.7 Coordination Assistant Member / Clubs Responsible Journey
- D.8 Admin Journey

E. Business Rules

TERMINOLOGY

Core domain concepts and platform definitions used throughout this specification:

- **Coordination Club ("BDE")**: The unique, non-deletable, school-level club bootstrapped at launch to manage other clubs.
- **System Role (Realm Role)**: A high-level platform identity assigned in Keycloak (`GUEST`, `USER`, `STUDENT`, `ADMIN`).
- **Club Role**: A role assigned to a user within a specific club's membership (`MEMBER`, `ASSISTANT_MEMBER`, `CLUB_PRESIDENT`).
- **Privilege (Grantable Endpoint)**: An active permission granted to an `ASSISTANT_MEMBER`, either directly as an individual endpoint or via a `ClubProfile`.
- **Category**: A functional grouping of grantable endpoints (`MANAGE_POSTS`, `MANAGE_EVENTS`, `MANAGE_MEMBERS`, `MANAGE_CLUBS`).
- **Club Profile**: A named custom preset belonging to a club that groups multiple grantable endpoints together (e.g., "Event Manager" profile).
- **Privileged Endpoint**: A special endpoint restricted exclusively to the Coordination Club.
- **Visibility**: Access control scope for posts and events (`PUBLIC`, `STUDENTS_ONLY`, `MEMBERS_ONLY`, `HIDDEN`).

A. SYSTEM AUTHORIZATION & GOVERNANCE

a. Roles Glossary

Role	Scope	Description
`ADMIN`	System	Platform administrator. Handles account bans, platform statistics, and system role changes.
`GUEST`	System	Unauthenticated viewer. Can browse public posts, events, and clubs.
`USER`	System	Authenticated user (unverified student). Can comment on and book public events.
`STUDENT`	System	Verified student account. Can view, comment on, and book student content, as well as request to join clubs.
`CLUB_PRESIDENT`	Club	Highest authority in a club. Has all privileges related to this club, including managing the privileges of assistant members.
`ASSISTANT_MEMBER`	Club	A club staff member having privileges granted dynamically by the `CLUB_PRESIDENT` (assigned directly or via `ClubProfile`).
`MEMBER`	Club	A member in a specific club. Can access members-only posts and events.

b. Role Hierarchy

Legend: *`X` → `Y` means `X` inherits all capabilities of `Y`.*

b.1 System Level:

`STUDENT` → `USER` → `GUEST`

b.2 Club Level:

`CLUB_PRESIDENT` → `ASSISTANT_MEMBER` → `MEMBER`

c. Visibility

Visibility Scope	Content Visible To	Commenting	Event Booking
`PUBLIC`	Everyone (Guest, User, Student)	`USER`+	`USER`+
`STUDENTS_ONLY`	`STUDENT`	`STUDENT`+	`STUDENT`+
`MEMBERS_ONLY`	`MEMBER` of the specific club	`MEMBER`+	`MEMBER`+
`HIDDEN`	`CLUB_PRESIDENT`/`ASSISTANT_MEMBER` with privileges	Disabled	Disabled

Note: Coordination Club Content is only `PUBLIC`, `STUDENTS_ONLY`, or `HIDDEN`. Every student is effectively a default recipient of BDE content without explicit membership.

e. Permissions

Dynamic Permissions Engine Structure

Permissions in the system are fully dynamic and driven by backend-registered API capabilities:

- **Grant Mechanics:**
  1. **Direct Endpoint Selection:** A President can assign specific endpoint names directly to an assistant member (e.g., `endpoints: ["create_post"]`).
  2. **Club Profile Preset:** A President can group endpoints into a named template (`ClubProfile`), then assign that profile to members (e.g., `EVENT_MANAGER`). Members inherit all endpoints configured in the profile.
- **Dynamic Propagation:** If a `ClubProfile`'s endpoint list is updated, all members assigned to that profile automatically acquire the updated capability set.
- **Source Attribution:** The system tracks whether a privilege was granted directly or originated from one or more `ClubProfile`s.
- **Privileged Scope Isolation:** Privileged endpoints are strictly prohibited from being granted in regular clubs and can only be assigned within the Coordination Club.

Situational Examples

Situation 1: Direct Endpoint Grant to a Club Assistant

Scenario: The President of the Robotics Club wants a member to publish posts without giving them event or member management permissions.
Action: The President assigns the individual endpoint `manage_posts` directly to the member.
Result: The member's role becomes `ASSISTANT_MEMBER` with the `manage_posts` capability. No reusable profile template is created.

Situation 2: Club Profile Preset Assignment

Scenario: The President of the Music Club wants a standard role for their event team.
Action: The President creates a `ClubProfile` named "Event Lead" containing endpoints `["create_event", "update_event", "manage_reservations"]`, then assigns "Event Lead" to two assistant members.
Result: Both members receive all three endpoints. Their privilege records link back to the "Event Lead" profile as the source.

Situation 3: Dynamic Propagation on Profile Template Update

Scenario: The Music Club President decides Event Leads should also be able to cancel events.
Action: The President updates the "Event Lead" `ClubProfile` to add `cancel_event`.
Result: Both assistant members assigned to "Event Lead" immediately gain the `cancel_event` capability.

Situation 4: Coordination Club

Scenario: The Coordination Club President delegates club management tasks to a member of his team.
Action: The Coordination President assigns the privileged endpoint `change_club_president` (`MANAGE_CLUBS` category) to the member.
Result: The member becomes a "Clubs Responsible" capable of reassigning club presidents across any school club.

Situation 5: System Admin Scope Boundaries

Scenario: A user violates platform policies.
Action: System `ADMIN` revokes the user's `STUDENT` status or bans their Keycloak account.
Boundary: The System `ADMIN` cannot publish club posts, edit club events, or manage internal club memberships.

B. PLATFORM & NAVIGATION

The system works across two platforms: the **Mobile Platform**, built for daily student use and club activities, and the **Web Platform**, which contains all mobile features along with tools for full school management and administration.

B.1 Mobile Platform

Designed for daily engagement and club-level tasks:

- `Home Feed`: Browse posts and events based on access level (`PUBLIC`, `STUDENTS_ONLY`, `MEMBERS_ONLY`), view attachments, and add comments.
- `Clubs Directory`: Explore clubs, view club leaders, and send join requests.
- `My Calendar`: Check booked events, see club activities, and cancel bookings.
- `User Profile`: View personal details, update profile picture, and view joined clubs.
- `Club Dashboard` (`CLUB_PRESIDENT` & `ASSISTANT_MEMBER`):
  - `Publishing`: Create, edit, and delete club posts and events with attachments.
  - `Member Management`: Accept or reject join requests, view members, and remove members.
  - `Permissions`: Create custom role profiles and give or remove permissions for assistant members.
  - `Statistics`: View club numbers and see lists of event attendees.

B.2 Web Platform

The complete version of the system that includes every mobile feature plus administrative tools:

- `Full Mobile Parity`: Complete access to feeds, event booking, commenting, club browsing, personal calendar, and content publishing.
- `Club Management`: Create new clubs, update club details, and assign or change club presidents.
- `User Management` (`ADMIN`): Verify student accounts (change `USER` to `STUDENT`) or ban accounts.
- `Coordination Club Setup` (`ADMIN`): Create the main coordination club and set its first president.
- `System Statistics` (`ADMIN`): View platform-wide numbers, including total users, club counts, and activity metrics.

C. SCREENS

Note: All screen mockups are available in the shared Google Drive:

https://drive.google.com/drive/folders/1drf6qMhRZ0CjouPv_Ow3bSoXTAtWCmRu?usp=sharing

D. PER-ROLE USER JOURNEYS

D.1 Guest to User Journey

1. **Explore Public Content**: A `GUEST` opens the app and browses `PUBLIC` posts, `PUBLIC` events, and the `Clubs Directory` without logging in.
2. **Sign-up Prompt**: When trying to book a `PUBLIC` event, leave a comment, or access the profile tab, the app prompts the `GUEST` to sign in or register.
3. **Account Creation**: The `USER` signs up with their basic details (name, username, email, and password) and logs in.
4. **Active Engagement**: Now logged in as a `USER`, they can:
   - Book spots for `PUBLIC` events and view them in their personal `My Calendar`.
   - Write comments on `PUBLIC` posts and events.
   - Edit `User Profile` details and update their profile picture.

D.2 Student Journey

1. **Explore Student Content**: A verified `STUDENT` logs in and browses `PUBLIC` and `STUDENTS_ONLY` posts and events on the `Home Feed`.
2. **Engage and Book**: The `STUDENT` comments on posts and events, and books spots for `STUDENTS_ONLY` events, which appear in their personal `My Calendar`.
3. **Browse and Request to Join**: The `STUDENT` browses the `Clubs Directory`, views club details and staff, and submits a join request to a club.
4. **Join the Club**: Once someone with `Member Management` privileges approves the request, the `STUDENT` becomes a `MEMBER` in that specific club, unlocking access to that club's `MEMBERS_ONLY` posts and events.

D.3 Club President Journey

1. **Take the Lead**: The `Coordination Club` assigns a `STUDENT` as `CLUB_PRESIDENT` when creating a new club or transferring leadership, unlocking full management access for that specific club.
2. **Set Up the Club**: The `CLUB_PRESIDENT` opens the `Club Dashboard`, updates the club description, uploads a club logo, and views the member list.
3. **Build the Team**: The `CLUB_PRESIDENT` creates custom permission sets (`ClubProfile`) and assigns them to selected members, promoting them to `ASSISTANT_MEMBER`s.
4. **Publish and Organize**: The `CLUB_PRESIDENT` posts club news and creates events, choosing whether content is visible to `PUBLIC`, `STUDENTS_ONLY`, or `MEMBERS_ONLY`.
5. **Manage Members**: The `CLUB_PRESIDENT` reviews pending join requests, accepts new members, and removes members when necessary.

D.4 Assistant Member Journey

1. **Receive Responsibilities**: A `MEMBER` is given specific permissions by the `CLUB_PRESIDENT` (directly or through a role profile) and becomes an `ASSISTANT_MEMBER`.
2. **Access the Dashboard**: The `ASSISTANT_MEMBER` can now open the `Club Dashboard`, seeing only the tools granted to them.
3. **Execute Assigned Duties**: Depending on their assigned permissions, the `ASSISTANT_MEMBER` can:
   - Create, update, or remove club posts and events.
   - View attendee lists for club events

D.5 Club Member Journey

1. **Welcome to the Club**: Once their join request is approved, the student officially becomes a `MEMBER` of that specific club.
2. **Access Exclusive Content**: The `MEMBER` gains immediate access to the club's `MEMBERS_ONLY` posts and events in their feed.
3. **Engage and Attend**: The `MEMBER` comments on `MEMBERS_ONLY` posts and events, and books spots for `MEMBERS_ONLY` events.
4. **Track Activities**: Booked club events automatically appear in the member's personal `My Calendar`, keeping them up to date with club schedules.

D.6 Coordination Club President Journey

1. **Appointment**: At the start of the school year, the `ADMIN` appoints a student as the `CLUB_PRESIDENT` of the `Coordination Club`, granting them leadership over the main school club.
2. **Build the Team**: The `CLUB_PRESIDENT` adds team members and prepares custom permission sets for the staff.
3. **Grant Special Permissions**: Along with standard permissions (like posting updates or creating events), the `CLUB_PRESIDENT` can grant unique, school-level privileges—such as creating clubs and managing club presidents—turning a staff member into a `CLUBS RESPONSIBLE`.
4. **Publish Campus Announcements**: The `CLUB_PRESIDENT` creates posts and events that reach the entire student body or the general public.

D.7 Coordination Assistant Member / Clubs Responsible Journey

1. **Receive Responsibilities**: A member of the `Coordination Club` is assigned special administrative permissions by their President, becoming a `CLUBS RESPONSIBLE`.
2. **Verify Students**: They review and verify student accounts, upgrading them from `USER` to `STUDENT` so they can access school-restricted features.
3. **Manage School Clubs**: With their club-management permissions, they create new clubs, update club details, and assign or change club presidents across the school.
4. **Handle Team Duties**: Depending on other permissions granted to them, they can also publish announcements and organize events for the `Coordination Club`.

D.8 Admin Journey

1. **Dedicated Access**: The `ADMIN` logs in with dedicated administrator credentials, operating strictly as a platform manager rather than a user.
2. **Set Up Coordination Leadership**: At the start of the school year, the `ADMIN` initializes the `Coordination Club` and appoints or changes its President.
3. **Account Security**: The `ADMIN` assists with account recovery and bans or removes accounts that violate school rules.

E. BUSINESS RULES

The system enforces a set of clear platform governance rules:

- **A student can be a `MEMBER` or `ASSISTANT_MEMBER` of multiple clubs, but `CLUB_PRESIDENT` of at most**